DFD-HR: Generalizable Deepfake Detection via Hierarchical Routing Learning

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Abstract

Developing generalizable deepfake detectors has become increasingly important with the rapid advancement of generative models. Adapting visual foundation models (VFMs), e.g. *CLIP, through parameter-efficient finetuning (PEFT), with only a small subset of parameters updated, has been proven highly effective for generalizable detection. However, the success of "fewer-parameters" training raises an important question: although only a few parameters are tuned, have existing PEFT-based detectors truly exploited the most informative ones while eliminating redundant parameters for better generalization? In this work, we move beyond standard PEFT by proposing a joint optimization strategy that operates at both the layer and token levels. Since latent features across layers capture different semantic abstractions and tokens within the same layer convey varied forgery cues, we propose integrating both layer-level and token-level routing to maximize representational synergy. Specifically, at the layer level, we introduce "Early Layer Pruning", an adaptive truncation mechanism that enables the model to adaptively learn distinct forward depths for different types of instances. At the token level, "Token Selection" is guided by the Spearman rank loss to filter tokens irrelevant to forgery learning, enabling the model to focus on the most discriminative cues. Furthermore, a unified MoE architecture is applied that encourages diversity and thus reduces the potential model's overfitting to specific forgery types. Extensive benchmarking results demonstrate the effectiveness of our designs and show the superior performance of our method over existing state-of-the-arts. Project: https://dfd-hr.github.io/.*

1. Introduction

DeepFake Detection (DFD) [2, 12, 39, 52, 58, 77] aims to protect facial privacy from digital-editing technology, *e.g.*, face-swapping [13, 35, 40], face-reenactment [31, 72, 73], and entire image synthesis [30]. However, as a real-time attack and defense task, deepfake detectors [63, 65, 71] suffer from overfitting to specific forgery artifacts, which severely hinders generalization for practical applications.

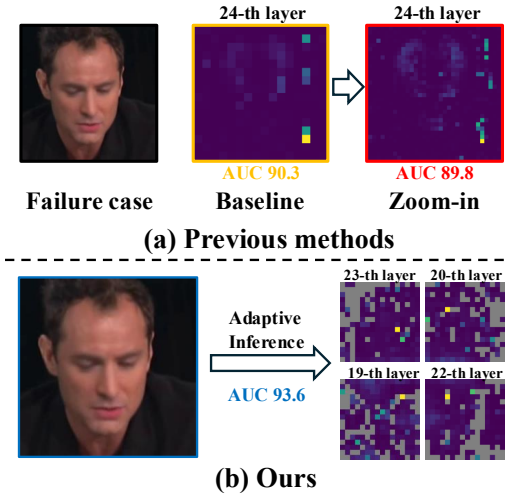

Figure 1. **Comparison of DFD-HR with previous methods.** Failure cases in forgery detection often exhibit no obvious artifacts at a glance. Attention is frequently drawn to incorrect regions even zoom-in. Our method overcome this by layer and token routing. Face images shown in figure are from Celeb-DF-v2 [40].

To address this issue, previous methods focus on the discovery of more general fake patterns, which can be divided into different directions: data augmentation [7, 37, 38, 65, 92], frequency clues [19, 45, 50, 58, 78], disentanglement learning [41, 43, 79, 86], reconstruction learning [75], and contrastive learning [11, 51]. Benefit from large-scale pretraining [21, 54, 57, 60, 89], researchers have attempted to leverage vision foundation models (VFMs) for DFD [11, 64, 83] through parameter-efficient finetuning (PEFT), achieving notable improvement in generalization.

However, despite their success, existing PEFT-based detectors still raise a fundamental question: *have these methods truly exploited the most informative parameters while eliminating redundant ones to achieve optimal generalization?* We argue that **simply updating a small number of parameters does not guarantee effective adaptation if the selected parameters or features are suboptimal**. In-

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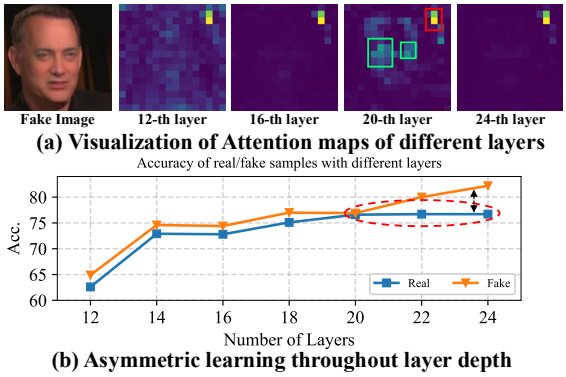


(a) Visualization of Attention maps of different layers

(b) Asymmetric learning throughout layer depth

Figure 2. **Illustration of importance for hierarchical routing.** Sub-figure (a) shows that the attention maps differ among layers and convey varied forgery cues influenced by spurious tokens. Sub-figure (b) indicates that real samples converge rapidly, while forged ones remain challenging and are progressively uncovered at deeper layers, proving the necessity of Early Layer Pruning. Face images shown in figure are from Celeb-DF-v2 [40].

deed, directly transferring CLIP [60] to DFD often results in insensitivity to subtle forgery cues that are easily overlooked, as shown in Fig. 1(a). "Zoom-in" strategies [48] partially alleviate this issue by focusing on local regions, but they still fail to capture forgery-related semantics consistently. Since latent features across layers capture different semantic abstractions (Fig. 2(a)), which are asymmetric needed for real and fake samples (Fig. 2(b)). In addition, tokens within the same layer convey varied forgery cues influenced by spurious tokens (red region in Fig. 2(a)).

To this end, we move beyond standard PEFT and introduce a hierarchical routing strategy operating at both the layer and token levels. **At the layer level**, we propose Early Layer Pruning. This adaptive truncation mechanism allows the model to learn distinct forward depths for different instances, where real samples typically terminate earlier while fake samples propagate deeper and differ across forgery methods (Fig. 5(a)). **At the token level**, we introduce Token Selection guided by the Spearman rank loss to filter out tokens irrelevant to forgery learning, enabling the model to focus on the most discriminative cues (Fig. 6). Moreover, we employ a **unified Mixture-of-Experts (MoE) architecture** that encourages diversity and thus helps to discriminate real and fake samples (Fig. 5(b)). By integrating layer- and token-level routing within a unified sparse architecture, the model dynamically learns to concentrate learning resources on the most informative representations.

To validate DFD-HR, we apply DFD-HR to three representative VFMs, *i.e.* BEITv2 [57], SigLIP [88] and CLIP [59], and conduct extensive experiments on the popular benchmarks. Experimental results show that DFD-HR can greatly exceed the SoTA methods, *e.g.* +2.3% average Video-level AUC gains on cross-dataset setting and +3.8% average Video-level AUC gains on cross-method setting. More importantly, we present a comprehensive analysis of hierarchical routing for its effectiveness of decoupling real and fake samples at both the layer and token levels.

Overall, the contributions of this paper are three-fold:

- We explicitly investigate the roles of parameters at both layer- and token- level for generalizable forgery learning.
- We present the first attempt of hierarchical routing network named DFD-HR with Early Layer Pruning, Token Selection and Expert Routing.
- DFD-HR achieves significant performance gains on cross-dataset and cross-method settings over SoTAs.

2. Related Work

Generalizable Deepfake Detectors. Deepfake detection (DFD) [2, 12, 39, 52, 58, 77] aims to detect synthetic media that can convincingly mimic the appearance and conduct of real people. Early DFD generalizable detectors [5, 37, 45, 50, 52, 53, 65, 81] mainly focus on discovering more general fake patterns, which can be roughly divided into four categories, *e.g.*, frequency-based methods [19, 45, 50, 58, 78], blend-based methods [65, 85], reconstruction-based methods [75, 90] and disentanglement-based methods [9, 43, 49, 79, 86]. However, these patterns suffer from limited generalization due to the rapid development of AI generative technologies [18, 22, 62, 84]. More recently, the remarkable transferability and generalization capabilities of large-scale pre-trained foundation models have attracted more attention. FFTG [70] proposes an accurate visual-linguistic dataset for the adaptation of CLIP [59] and MLLMs [48]. VLF-FFD [56] uses a novel external detector fusion solution for MLLM-enhanced face forgery detection. DeepShield [4] balances local sensitivity and global generalization to improve robustness on unseen forgeries.

Parameter Efficient Modeling. Parameter Efficient Fine-tuning (PEFT) of Vision Transformer (ViT) has made great progress in transfer learning [20, 47, 68], and can be broadly categorized into three approaches, *i.e.* prompt-tuning [6, 32, 42], Adapter-tuning [11, 34, 64, 85], and LoRA-based methods [83, 91]. In particular, Forensics Adapter [11] designs an adapter for CLIP to learn the unique blending boundaries of forged faces. Effort [83] employs SVD to LoRA [24] to construct two orthogonal subspaces of foundation models for efficient fake learning. More recently, sparsely-gated Mixture of Experts networks (MoEs) have demonstrated excellent scalability in Natural Language Processing [26] and Computer Vision [61]. MoE-FFD [34] updates lightweight mixture of LoRA and Adapter layers to scale the model's capacity and smartly select optimal forgery experts. More recently, token-level methods [55, 76], such as UDD [17], overcome position bias and content bias via token-level shuffling and mixing.

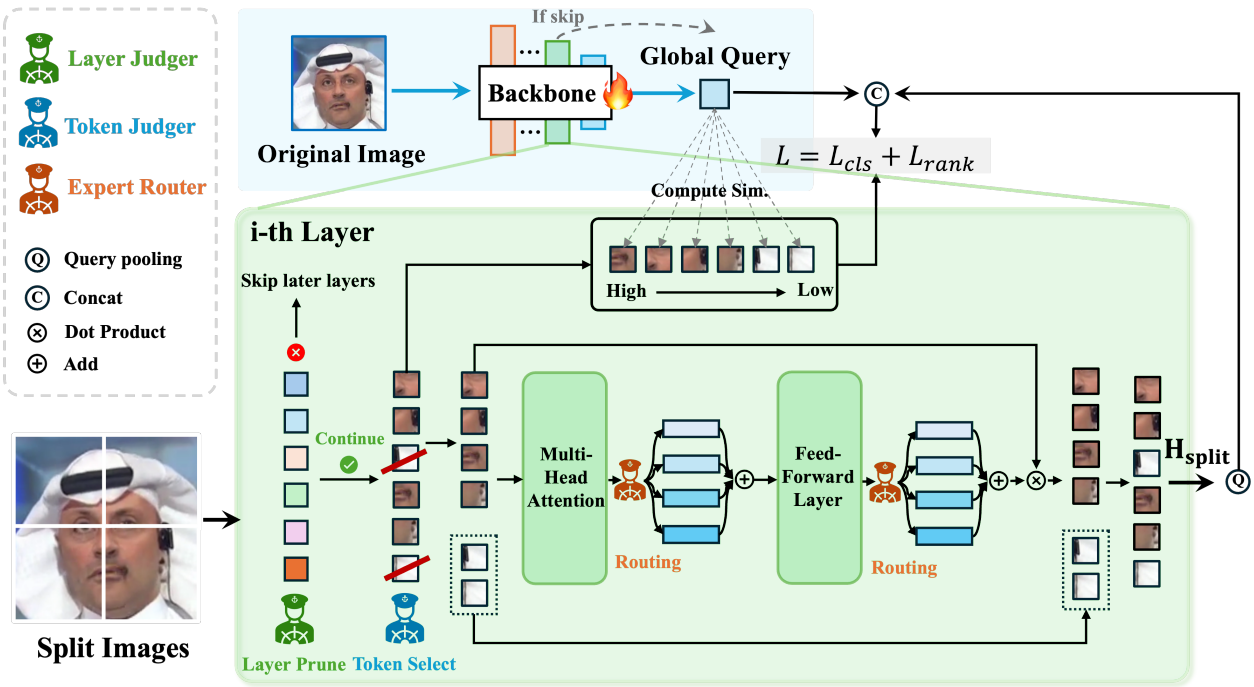


Figure 3. **The overall framework of DFD-HR.** DFD-HR explicitly exploits the most informative parameters via Hierarchical Routing. At the layer level, Early Layer Pruning enables the model to adaptively learn distinct forward depths for different instances. At the token level, Token Selection is guided by the Spearman rank loss to filter tokens irrelevant to forgery learning. Moreover, Expert Routing encourages diverse expert activation, which helps to discriminate real and fake samples. Face images shown in figure are from Celeb-DF-v2 [40].

3. Method

The overall pipeline of the proposed approach is illustrated in Fig. 3, designed to explicitly investigate the roles of parameters via hierarchical routing. Specifically, at the layer level, "Early Layer Pruning" is an adaptive truncation mechanism that enables distinct forward depths among instances. At the token level, "Token Selection" is used to filter forgery-irrelevant tokens and "Expert Routing" further gives differentiated weights between real and fake samples, thereby improving generalizable detection performance.

3.1. Hierarchical Routing (HR)

We first propose Hierarchical Routing to tackle the insensitivity to subtle forgery artifacts of foundation models. The most straightforward approach is to apply the "Zoom-in" strategies in Multi-Modal Large Language Model [48]. However, directly transferring this paradigm to DFD partially alleviate this issue by focusing on local regions, but still fail to capture forgery-related semantics consistently. Therefore, we divide the original image into splits and leverage global query information to help the model attend to facial forgeries. Given a set of labeled data $D = \{(\mathcal{I}_i), \mathcal{Y}_i\}_{i=1}^{N}$, the target of HR can be formulated as

$$\begin{aligned} \mathcal{I}^s &= \mathrm{Split}(\mathrm{Upsample}(\mathcal{I})), \\ \min &\mathcal{L}(\theta; \{(\mathcal{I}, \mathcal{I}^s), \mathcal{Y}\}). \end{aligned} \tag{1}$$

Here, $\mathcal{L}$ is the objective function. θ denotes HR proposed to exploit the most informative parameters. $\mathcal{I}$, $\mathcal{I}^s$ and $\mathcal{Y}$ denote images, split images and labels, respectively.

3.1.1. Early Layer Pruning

To distinguish among instances, we first explore the relationships between layers. Considering that the latent features across layers capture different semantic abstractions (Fig. 2(a)) which are asymmetric needed for real and fake samples (Fig. 2(b)), it suggests that layer routing can be optimized for better interactions. Based on these observations, we further propose an adaptive truncation for layers.

Given input hidden states $\mathbf{H}_i \in \mathbb{R}^{B\times L\times D}$ of i-th layer, we first pass [CLS] token $h_i^{cls} \in \mathbb{R}^{B\times D}$ to layer judge:

$$\mathrm{logits} = \mathbf{W_2}(\mathrm{ReLU}(\mathbf{W_1}(h_i^{cls}))), \tag{2}$$

where $\mathrm{W}_1 \in \mathbb{R}^{D\times d}$ and $\mathrm{W}_2 \in \mathbb{R}^{d\times 1}$ are learnable weights. Then, we use Gumbel-Sigmoid [27] to perform hard routing during training. First, the Gumbel noise distribution is converted from uniform distribution sampling:

$$\begin{aligned} U &\sim \mathrm{Uniform}(0,1), \\ g &= -\log(-\log(U+\epsilon)+\epsilon), \end{aligned} \tag{3}$$

where U and g are the uniform and Gumbel noise distribution variables, respectively. ϵ is a numerical stability con-

stant. The corresponding probability can be formulated as:

$$p_{\text{soft}} = \sigma\left(\frac{\text{logits} + g}{\tau}\right), \quad p_{\text{hard}} = \mathbb{I}\left[p_{\text{soft}} > 0.5\right], \tag{4}$$

where $\sigma(\cdot)$ is Sigmoid function, τ is a temperature parameter. The soft routing probability $p_{\text{soft}} \in [0, 1]$ and the corresponding hard one $p_{\text{hard}} \in \{0, 1\}$.

During training, the output probs p is then defined as:

$$p = \text{sg}\left(p_{\text{hard}}\right) - \text{sg}\left(p_{\text{soft}}\right) + p_{\text{soft}}, \tag{5}$$

where $\text{sg}(\cdot)$ is stop-gradient function and the gradient backpropagation is from p_{soft} during training. And p_{hard} is used directly during inference. The output hidden states $\mathbf{H}_{i+1} \in \mathbb{R}^{B\times L\times D}$ of i-th layer can be formulated as:

$$\mathbf{H}_{i+1} = p \cdot \text{Layer}_i\left(\mathbf{H}_i\right) + (1 - p) \cdot \mathbf{H}_i. \tag{6}$$

To encourage the difference between instances, once $p = 0$ in i-th layer, that instance will not participate in all subsequent layers and is directly used for classification.

3.1.2. Token Selection

Previous work [17] shows vision foundation models exhibit semantic bias in DFD. We attribute it to the fact that tokens within the same layer convey varied forgery cues influenced by spurious tokens (Fig. 2(a)). Based on the observation, "Token Selection" is proposed to filter tokens irrelevant to forgery learning via global query prior.

To distinguish the importance of each token for forgery detection, we first introduce token judge to score all tokens:

$$\text{scores}_i = \mathbf{W}_{\text{t}}([h_i^{cls}, h_i^{patch}]), \tag{7}$$

where $\text{scores}_i \in \mathbb{R}^{B\times L}$ is the forgery scores of all tokens in i-th layer, and $\mathbf{W}_{\text{t}} \in \mathbb{R}^{D\times 1}$ is the learnable weight. For each layer, only the TopK score tokens will be selected:

$$\begin{aligned} k &= \max(1, \lfloor r \times L \rfloor), \\ \mathbf{H}_i^{select} &= \mathbf{H}_i[\text{TopK}\left(\text{scores}_i, k\right)], \end{aligned} \tag{8}$$

where r and k are the proportion and number of selected tokens, L is the number of all tokens in i-th layer, and $\mathbf{H}_i^{select} \in \mathbb{R}^{B\times k\times D}$ is the selected tokens scored by token judge. Then, filtered tokens $\mathbf{H}_i^{filter} \in \mathbb{R}^{B\times (L-k)\times D}$ are directly propagated to the output of i-th layer.

Based on the prior knowledge that forgery cues appear only in face regions, We impose a rank constraint on the token judge to better guide token selection. Given the final output [CLS] token h_{global}^{cls} of the image $\mathcal{I}$ which represents global face information, we calculate cosine similarity between h_{global}^{cls} and input hidden states H_i of split images $\mathcal{I}^s$:

$$\text{cosine}_{\text{i}} = \frac{\mathbf{H}_i}{\|\mathbf{H}_i\|_2} \cdot \left(\frac{h_{global}^{cls}}{\|h_{global}^{cls}\|_2}\right)^{\text{T}}, \tag{9}$$

where $\|\cdot\|_2$ denotes the L2 norm, $\text{cosine}_{\text{i}} \in \mathbb{R}^{B\times L}$ denotes the output similarity obtained by the product of $\mathbf{H}_i \in \mathbb{R}^{B\times L\times D}$ and $h_{global}^{cls} \in \mathbb{R}^{B\times 1\times D}$. To encourage the monotonic relationship between scores_i and cosine_{i}, we define rank loss using the Spearman correlation coefficient [66]:

$$\begin{aligned} \rho_{\text{spearman}} &= \frac{\text{Cov}(\text{rank}(\text{scores}_{\text{i}}), \text{rank}(\text{cosine}_{\text{i}}))}{\sigma_{\text{rank}(\text{scores}_{\text{i}})} \cdot \sigma_{\text{rank}(\text{cosine}_{\text{i}})}}, \\ \mathcal{L}_{rank} &= 1 - \rho_{\text{spearman}}. \end{aligned} \tag{10}$$

Here, $\text{rank}(\cdot)$ calculates token ranks according to corresponding metrics, $\text{Cov}(\cdot, \cdot)$ is the covariance of two ranked sequences, and $\sigma_{(\cdot)}$ is the standard deviation function. The output $\rho_{\text{spearman}} \in [-1, 1]$ and $\mathcal{L}_{rank}$ is optimized for consistent correlation. Equipped with the well-learned token judge, the selected tokens $\mathbf{H}_i^{select}$ are passed to Transformer block Layer_i to get the output $\mathbf{H}_{i+1}^{select}$. Then, the output are reweighted according to scores of selected tokens:

$$\begin{aligned} \text{scores}_i^{select} &= \text{Softmax}(\text{TopK}\left(\text{scores}_i, k\right)), \\ \mathbf{H}_{i+1}^{select} &= \mathbf{H}_{i+1}^{select} \odot (1 + \text{scores}_i^{select}), \end{aligned} \tag{11}$$

where Layer_i consists of Multi-Head Attention (MHA) and MLP function in Eq. (12). $\text{scores}_i^{select} \in \mathbb{R}^{B\times k}$ denotes scores of selected tokens and $\mathbf{H}_{i+1}^{select} \in \mathbb{R}^{B\times k\times D}$ is the final output of selected tokens. Then, the final output $\mathbf{H}_{i+1}$ of i-th layer is the merge of $\mathbf{H}_{i+1}^{select}$ and filtered tokens $\mathbf{H}_i^{filter}$, maintaining their original relative order.

3.1.3. Token-level Expert Routing

To mitigate conflicts of forgery learning of different types in the i-th layer, we insert Adapter after Multi-Head Attention (MHA) and MLP via expert routing. Given selected token $\mathbf{H}_i^{select}$ ($\mathbf{X}$ for simplify), Layer_i can be formulated as:

$$\begin{aligned} \mathbf{X}_{\text{attn}} &= \mathbf{X} + \text{MOA}(\text{Attention}(\text{LN}_1(\mathbf{X}))), \\ \mathbf{X}_{\text{out}} &= \mathbf{X}_{\text{attn}} + \text{MOA}(\text{MLP}(\text{LN}_2(\mathbf{X}_{\text{attn}}))). \end{aligned} \tag{12}$$

Here, $\text{MOA}(\cdot)$ is our Mixture of Adapter design, which is composed of gate network with noise, Adapter Tuning and routing mixture. First, gate network can be formulated as:

$$\begin{aligned} \mathbf{L} &= \mathbf{X}\mathbf{W}_g, \\ \mathbf{G} &= \text{Softmax}(\mathbf{L} + \boldsymbol{\epsilon} \odot \text{Softplus}(\mathbf{L})), \end{aligned} \tag{13}$$

where N is the expert number and $\mathbf{W}_g \in \mathbb{R}^{D\times N}$ denotes the learnable gating parameter. $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ is added only during training and $\mathbf{G} \in \mathbb{R}^{B\times k\times N}$ denotes gating weights for each selected tokens. The final output of MOA is:

$$\begin{aligned} \text{Adapter}(\mathbf{X}) &= \mathbf{W_2}(\text{ReLU}(\mathbf{W_1}(\mathbf{X})), \\ \text{MOA}(\mathbf{X}) &= \mathbf{X} + \sum_{i=1}^{N} \mathbf{G}_i \odot \text{Adapter}_i(\mathbf{X}). \end{aligned} \tag{14}$$

3.2. Multi-Scale Fusion (MSF)

Following D^3 [87], images at multiple scales exhibit distinct representations that complement one another. This motivates us to fuse between split images and original images, jointly learning of shared forgery artifacts.

$$\mathbf{h}_{split} = \mathrm{MHA}(\mathbf{Q}_{\text{token}}, \mathbf{H}_{\text{split}}, \mathbf{H}_{\text{split}}), \tag{15}$$

where $\mathbf{Q}_{\text{token}} \in \mathbb{R}^{B\times 1\times D}$ is the learnable query token, $\mathbf{H}_{\text{split}} \in \mathbb{R}^{B\times n\times D}$ denotes the output hidden states of split images, and n is the number of splits per image. Finally, $\mathbf{h}_{split}$ and $\mathbf{h}^{cls}_{global}$ are concatenated for classification.

Overall, the complete procedure of DFD-HR is described in Algorithm 1. The training objective combines the classification loss (*i.e.*, cross-entropy loss for binary classification) and rank loss (*i.e.*, the Spearman correlation coefficient for better Token Selection):

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_1 \frac{1}{m} \sum_{i}^{m} \mathcal{L}^{i}_{\text{rank}}. \tag{16}$$

where λ_1 is hyperparameter that balances the importance of the regularization term, and m represents the number of Hierarchical Routing inserted into the branch.

Algorithm 1 *DFD-HR* Approach Algorithm

Input: Training data $\mathcal{D} = \{\mathcal{I}, \mathcal{Y}\}$; Pretrained CLIP model θ; Total block number M; HR inserted block number m; Token Selection ratio r; Hyperparameter λ_1
Output: Layer judge $\{\theta^i_l\}_{i=M-m}^{M-1}$; Token judge $\{\theta^i_t\}_{i=M-m}^{M-1}$; Expert network $\{\theta^i_e\}_{i=1}^{M}$

▶ **Training Loop**
for each epoch **do**
  for each batch in $\mathcal{D}$ **do**
    ▷ **Original Image Forward Pass**
    ◇ Same as Split Images Forward Pass shown bellow
    ◇ Extract h^{cls}_{global} of last hidden states
    ▷ **Split Images Forward Pass**
    ◇ Split images by Eq. (1)
    for each Layer_i **do**
      if $i >= M - m$ **then**
        ◇ Forward Layer judge by Eq. (2)
        ◇ Early Layer Pruning by Eq. (6)
        if $p = 0$ **then**
          Break
        end if
        ◇ Forward Token judge by Eq. (7)
        ◇ Compute $\mathcal{L}_{\text{rank}}$ using h^{cls}_{global} by Eq. (10)
        ◇ Token Selection by Eq. (8)
      end if
      ◇ Forward Expert network by Eq. (12)
    end for
    ▷ **Multi-Scale Fusion**
    ◇ Fuse the final output by Eq. (15)
    ▷ **Total Loss**
    ◇ Compute classification loss $\mathcal{L}_{cls}$
    ◇ $\mathcal{L} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{\text{rank}}$
    ▷ **Backward Pass and Optimization**
    ◇ Update $\theta_l, \theta_t, \theta_e$ using gradient descent
  end for
end for
return Updated weight of $\theta_l, \theta_t, \theta_e$

4. Experiment

4.1. Datasets and Metric

Following previous works [11, 82, 83], we adopt two widely used protocols for evaluation: **Protocol-1:** cross-dataset evaluation and **Protocol-2:** cross-manipulation evaluation within the FF++ data domain. For **Protocol-1**, we conduct evaluations by training the models on FaceForensics++ (FF++) [63] and testing them on other seven deepfake detection datasets: Celeb-DF-v2 (CDF-v2) [40], DeepfakeDetection (DFD) [15], Deepfake Detection Challenge (DFDC) [14], the preview version of DFDC (DFDCP) [16], DeeperForensics (DFo) [28], WildDeepfake (WDF) [94], and FFIW [93]. Note that FF++ has three different compression versions and we adopt the c23 version for training all methods following most existing works [81]. For **Protocol-2**, we evaluate the models on the latest deepfake dataset DF40 [82], which contains the forgery data generated within the FF++ domain, ensuring the fake methods different while the data domains remain unchanged. For fair comparison with other works [11, 44, 65, 82, 83], both **Video-level and Frame-level Area Under the Curve (AUC)** are reported, which measure the quality of deepfake detectors. We compute the average model's output probabilities of each video to obtain the video-level AUC.

4.2. Experimental Settings

Following the pre-processing and training pipeline of DeepfakeBench [80], we utilize CLIP ViT-L/14 [59] as our default vision backbone. We use full finetuning with learning rate of $1e-6$ as our baseline. Additionally, following [65], we sample 8 frames from each video for training with 16 batch size and 32 frames for inference with 32 batch size. DFD-HR is optimized by Adam [33] using the fixed learning rate of $1e-4$. We also use data augmentations following Effort [83]. For better performance, We insert HR into the last 4 layer of the backbone and equip other layers with expert designs. The ratio of token selection is set to 75% and λ_1 is 0.1. The expert number for each MOA is 4 and the hidden dimension of Adapter is set to 256.

4.3. Experimental Results

4.3.1. Generalization Evaluation

Comparison with competitive detectors. We first compare our DFD-HR with state-of-the-arts on video-level AUC in Tab. 1. DFD-HR obtains obvious gains on all settings, *i.e.*, +2.3% on cross-dataset and +3.8% on cross-

Table 1. **Benchmarking Results of Cross-dataset Evaluations (Protocol-1) and Cross-method Evaluations (Protocol-2)**. All detectors are trained on FF++_c23 [63] and evaluated on other fake data **(Video-level AUC)**. All the results are cited from [10, 80, 81, 83].

Methods	Cross-dataset Evaluation								Cross-method Evaluation								
	CDF-v2	DFD	DFDC	DFDCP	DFo	WDF	FFIW	Avg.	UniFace	BleFace	MobSwap	e4s	FaceDan	FSGAN	InSwap	SimSwap	Avg.
F3Net [58]	0.789	0.844	0.718	0.749	0.730	0.728	0.649	0.743	0.809	0.808	0.867	0.494	0.717	0.845	0.757	0.674	0.746
SPSL [45]	0.799	0.871	0.724	0.770	0.723	0.702	0.794	0.769	0.747	0.748	0.885	0.514	0.666	0.812	0.643	0.665	0.710
SRM [50]	0.840	0.885	0.695	0.728	0.722	0.702	0.794	0.767	0.749	0.704	0.779	0.704	0.659	0.772	0.793	0.694	0.732
CORE [53]	0.809	0.882	0.721	0.720	0.765	0.724	0.710	0.762	0.871	0.843	0.959	0.679	0.774	0.958	0.855	0.724	0.833
RECCE [5]	0.823	0.891	0.696	0.734	0.784	0.756	0.711	0.779	0.898	0.832	0.925	0.683	0.848	0.949	0.848	0.768	0.844
SLADD [7]	0.837	0.904	0.772	0.756	0.800	0.690	0.683	0.777	0.878	0.882	0.954	0.765	0.825	0.943	0.879	0.794	0.865
SBI [65]	0.886	0.827	0.717	0.848	0.899	0.703	0.866	0.821	0.724	0.891	0.952	0.750	0.594	0.803	0.712	0.701	0.766
UCF [79]	0.837	0.867	0.742	0.770	0.808	0.774	0.697	0.785	0.831	0.827	0.950	0.731	0.862	0.937	0.809	0.647	0.824
IID [25]	0.838	0.939	0.700	0.689	0.808	0.666	0.762	0.789	0.839	0.789	0.888	0.766	0.844	0.927	0.789	0.644	0.811
LSDA [81]	0.875	0.881	0.701	0.812	0.768	0.797	0.724	0.794	0.872	0.875	0.930	0.694	0.721	0.939	0.855	0.793	0.835
ProDet [10]	0.926	0.901	0.707	0.828	0.879	0.781	0.751	0.828	0.908	0.929	0.975	0.771	0.747	0.928	0.837	0.844	0.867
CDFA [44]	0.938	0.954	0.830	0.881	0.973	0.796	0.777	0.878	0.762	0.756	0.823	0.631	0.803	0.942	0.772	0.757	0.781
Effort [83]	0.956	0.965	0.843	**0.909**	0.977	0.848	0.921	0.917	0.962	0.873	0.953	0.983	0.926	0.957	0.936	0.926	0.940
DFD-HR (Ours)	**0.960**	**0.980**	**0.865**	0.901	**0.997**	**0.907**	**0.968**	**0.940**	**0.988**	**0.956**	**0.986**	**0.993**	**0.970**	**0.984**	**0.981**	**0.970**	**0.978**

Table 2. **Cross-dataset evaluation results (Frame-level AUC)**. All methods are trained on FF++ and evaluated on other datasets. All the results are cited from [11]. The **best** results are indicated in bold and the second-best results are underlined.

Method	Venue	CDF-v2	DFDC	DFD	WDF	Avg.
Xception [63]	ICCV'19	0.737	0.708	0.816	0.662	0.731
EfficientB4 [71]	ICML'19	0.749	0.696	0.815	0.610	0.718
F3Net [58]	AAAI'20	0.735	0.702	0.798	0.677	0.728
X-ray [37]	CVPR'20	0.679	0.633	0.766	-	-
FFD [12]	CVPR'20	0.744	0.703	0.802	-	-
SPSL [45]	CVPR'21	0.765	0.704	0.812	-	-
SRM [50]	CVPR'21	0.755	0.700	0.812	-	-
Recce [5]	CVPR'22	0.732	0.713	0.812	-	-
SBI [65]	CVPR'22	0.813	-	0.774	0.672	-
UCF [79]	ICCV'23	0.753	0.719	0.807	-	-
ED [1]	AAAI'24	0.864	0.721	-	-	-
CFM [46]	TIFS'24	0.828	-	0.915	0.784	-
FoCus [74]	TIFS'24	0.720	0.669	-	0.722	-
LSDA [81]	CVPR'24	0.830	0.736	0.880	-	-
DiffusionFake [69]	NeurIPS'24	0.805	-	0.904	0.801	-
MoE-FFD [34]	ArXiV'25	0.867	-	0.904	0.806	-
Dual-Adapter [64]	IJCV'25	0.717	0.727	-	-	-
UDD [17]	AAAI'25	0.869	0.758	0.910	-	-
Effort [83]	ICML'25	0.901	0.798	0.923	-	-
FFTG [70]	CVPR'25	0.832	-	0.948	0.851	-
ForAda [11]	CVPR'25	0.900	**0.843**	0.933	0.803	0.870
DFD-HR (Ours)	-	**0.910**	**0.843**	**0.953**	**0.856**	**0.891**

method evaluation compared to Effort [83]. We further compare DFD-HR with other methods on frame-level AUC in Tab. 2. DFD-HR also performs best on all four datasets and gains +2.1% average AUC over ForAda [11]. Besides, DFD-HR outperforms existing Adapter-based methods (MoE-FFD [34], Dual-Adapter [64], Effort [83], ForAda [11]) and token-level methods (UDD [17]). More importantly, DFD-HR is the first hierarchical routing model, *i.e.*, the number of layers is reduced by up to 17%, where the number of tokens is reduced by 25%.

4.3.2. Ablation Study

The impact of Hierarchical Routing and Multi-Scale Fusion. In Table 3, we first ablate Hierarchical Routing (HR) on cross-method evaluation. We can see that the routing metrics of both layer and token level can bring obvious performance gains on DFD datasets, *i.e.*, + 3.2% on CDF-v2 and + 1.9% on FaceShifter. Such gains also validate the necessity of hierarchical routing via explicit modeling. Then, Multi-Scale Fusion can also improve 0.7% average video-level AUC, further learning shared forgery artifacts.

Table 3. **Ablation studies on Hierarchical Routing (HR) and Multi-Scale Fusion (MSF) using Video-level AUC (%)**.

Ours		CDF-v2	FaceShifter	Avg.
HR	MSF			
×	×	91.8$_{+0.0}$	88.8$_{+0.0}$	90.3$_{+0.0}$
✓	×	95.0$_{+3.2}$	90.7$_{+1.9}$	92.9$_{+2.6}$
×	✓	92.9$_{+1.1}$	90.1$_{+1.3}$	91.5$_{+1.2}$
✓	✓	**96.0**$_{+4.2}$	**91.2**$_{+2.4}$	**93.6**$_{+3.3}$

Table 4. **Ablation studies on Early Layer Pruning, Token Selection and Expert Routing using Video-level AUC (%).**

Hierarchical Routing			CDF-v2	FaceShifter	Avg.
ELP	TS	ER			
×	×	×	91.8$_{+0.0}$	88.8$_{+0.0}$	90.3$_{+0.0}$
✓	×	×	93.4$_{+1.6}$	89.2$_{+0.4}$	91.3$_{+1.0}$
✓	✓	×	94.3$_{+2.5}$	90.4$_{+1.6}$	92.4$_{+2.1}$
✓	✓	✓	**95.0**$_{+3.2}$	**90.7**$_{+1.9}$	**92.9**$_{+2.6}$

The impact of components in Hierarchical Routing. In Tab. 4, we ablate the effectiveness of components in HR. At the layer level, Early Layer Pruning brings a significant improvement compared to our baseline, *e.g.*, +1.6% video-level AUC on CDF-v2. Early Layer Tuning learns distinct forward depths for different instances for better interactions. Moreover, at the token level, Token Selection further gives +1.1% improvement on average video-level AUC, suggesting that spurious tokens are filtered, enabling the model to focus on the most discriminative cues. Lastly, Expert Routing decouples forgery learning between real and fake samples, *e.g.*, +0.7% video AUC on CDF-v2. These results well validate the effectiveness of components in HR.

Comparison of Token Selection with other designs. Token Selection is compared with other popular designs in Tab. 5. PCA is a feature selection strategy used in LongCLIP [89]. FastV [8] is a popular token selection metric in

Table 5. **Comparison of methods for Token Selection on video-level AUC (%).** Our settings are in blue.

Settings	CDF-v2	FaceShifter	Avg.
Early Layer Pruning	93.4$_{+0.0}$	89.2$_{+0.0}$	91.3$_{+0.0}$
PCA [89]	93.3$_{-0.1}$	90.0$_{+0.8}$	91.7$_{+0.4}$
FastV [8]	93.5$_{+0.1}$	90.1$_{+0.9}$	91.8$_{+0.5}$
Sparse Attention [67]	92.8$_{-0.6}$	89.2$_{+0.0}$	91.0$_{-0.3}$
Token Selection	**94.3**$_{+0.9}$	**90.4**$_{+1.2}$	**92.4**$_{+1.1}$

Table 6. **Comparison of methods for multi-scale complementarity on video-level AUC (%).** Our settings are in blue.

Settings	CDF-v2	FaceShifter	Avg.
Baseline	91.8$_{+0.0}$	88.8$_{+0.0}$	90.3$_{+0.0}$
Split Ensemble	90.8$_{-1.0}$	89.9$_{+1.1}$	90.4$_{+0.1}$
Multi CLS Pooling	91.4$_{-0.4}$	86.5$_{-2.3}$	89.0$_{-1.3}$
D^3 [87]	89.4$_{-2.4}$	88.7$_{-0.1}$	89.1$_{-1.2}$
RINE [36]	83.9$_{-7.9}$	84.4$_{-4.4}$	84.2$_{-6.1}$
MRA [48]	91.4$_{-0.4}$	89.8$_{+1.0}$	90.6$_{+0.3}$
Multi-Scale Fusion	**92.9**$_{+1.1}$	**90.1**$_{+1.3}$	**91.5**$_{+1.2}$

MLLM based on attention weights. Sparse Attention [67] is a sparse and discrete manner searching for non-semantic features used in image manipulation localization. Both approaches can select forgery artifacts but our approach performs better, *e.g.*, +1.0% AUC on CDF-v2 compared to PCA and +1.4% average AUC compared to SparseViT.

Comparison of Multi-Scale Fusion with other designs. Multi-Scale Fusion is also compared with other viable choices. An ensemble of original images and split images is a direct solution during training and inference. Other classic methods such as RINE [36] and Multi CLS Pooling use [CLS] tokens of all layers for multi-scale information. D^3 [87] uses patch shuffling to focus on local forgery. MRA [48] introduces DINOv2 [27] for high-resolution fusion. Our Multi-Scale Fusion is both simple and effective, *e.g.*, +1.5% AUC on CDF-v2 compared to MRA.

Comparison of different PEFT methods. We further compare our proposed DFD-HR with other popular parameter-efficient finetuning (PEFT) methods (Supp.). The first observation is that Adapter [23] and LoRA [24] are better among PEFT variants. Then, LoRA variants like rsLoRA [29] and OLoRA [3] are better than traditional LoRA. Moreover, our proposed DFD-HR outperforms all other PEFT methods on the DFD benchmark.

Compatibility with other vision foundation models. In Tab. 7, following Effort [83], we transfer our multi-level routing method to other visual backbones, *i.e.*, BEIT-v2 [57] and SigLIP [88]. These results show that our method is compatible with various backbones for better generalization, *i.e.*, +4.7% average AUC with BEIT-v2 and +3.3% average AUC with SigLIP compared to Effort.

Table 7. **Compatibility with other vision foundation models** trained on FF++ (c23) and tested on CDF-v2 and SimSwap.

VFMs	#ImgSize	CDF-v2	SimSwap	Avg.
BEIT-v2 [57]	224	0.855	0.821	0.838
Effort [83]	224	0.894	0.850	0.872
Ours	224	**0.912**	**0.925**	**0.919**
SigLIP [88]	256	0.877	0.713	0.795
Effort [83]	256	0.895	0.778	0.867
Ours	256	**0.907**	**0.892**	**0.900**
CLIP [59]	224	0.857	0.860	0.859
Effort [83]	224	0.956	0.926	0.941
Ours	224	**0.960**	**0.970**	**0.965**

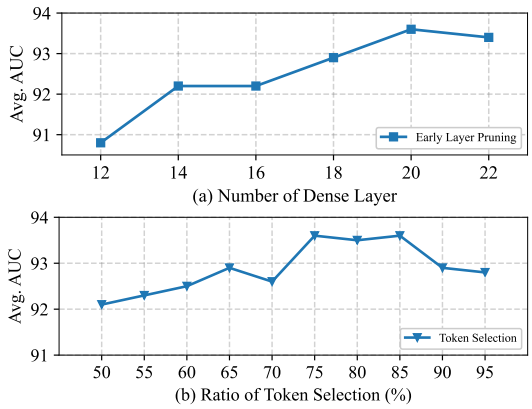


Figure 4. **Hyperparameter of Early Layer Pruning and Token Selection.** The default setting of the number of dense layer $M-m$ and selection ratio are set to 20 and 75% respectively.

Detailed ablations of proposed designs. We ablate detailed designs in our DFD-HR. **For Multi-Scale Fusion**, compared to mean or self-attention pooling, a learnable query aggregates local information of split images better (Tab. 8a). **For Early Layer Pruning**, the layer judges are Gumbel-Sigmoid hard routers on [CLS] token during training, different across layers for better performance (Tab. 8b). **For Token Selection**, the learnable token judge constraint by spearman rank loss is better than using cosine similarity weight to ensemble the score (Tab. 8c). Besides, [CLS] and patch tokens can be treated equally for selection (Supp.). The reweighting metric performed in front of residual of MLP is worse (Supp.). Token judge shared across layers performs worse (Supp.). **For Token-level Expert Routing**, expert design with gate noise during training is better than Top2 hard routing or balance loss constraint (Tab. 8d). **For Hyperparameters**, the more experts will not be better (Tab. 8e), λ_1 set to 0.1 is better (Tab. 8f), the number of sparse layers m and token selection ratio set to 4 and 75% respectively are better (Fig. 4).

4.3.3. Further Analysis

Analysis of Early Layer Pruning. We first analyze the distribution of Early Layer Pruning (Fig. 5 (a)). The first observation is that all distributions asymptotically assume a

case	CDF.	Fsh.	Avg.
Mean-pooling	95.7	91.1	93.4
SA-pooling	92.3	90.4	91.4
Query-select	**96.0**	**91.2**	**93.6**

(a) **Fusion metrics**. A learnable query can aggregate local information better.

case	CDF.	Fsh.	Avg.
Early Pruning	96.0	**91.2**	**93.6**
w.o. Gumbel	80.8	68.7	74.8
w.i. Shared	95.3	88.2	91.8
w.i. GAP	**96.1**	88.6	92.4

(b) **Layer Pruning metrics**. Layer judges differ across layers with Gumbel on [CLS].

case	CDF.	Fsh.	Avg.
Random	94.8	90.8	92.8
Learnable	95.4	91.0	93.2
Learn.+Weight	95.6	90.9	93.3
Learn.+Rank	**96.0**	**91.2**	**93.6**

(c) **Token Selection metrics**. Spearman Rank loss on the learnable token judges is better.

case	CDF.	Fsh.	Avg.
MoE	95.7	91.0	93.4
MoE+Top2	95.9	89.8	92.9
MoE+Balance	94.2	90.5	92.4
MoE+Noise	**96.0**	**91.2**	**93.6**

(d) **Expert Routing designs**. Experts with noise gate perform better.

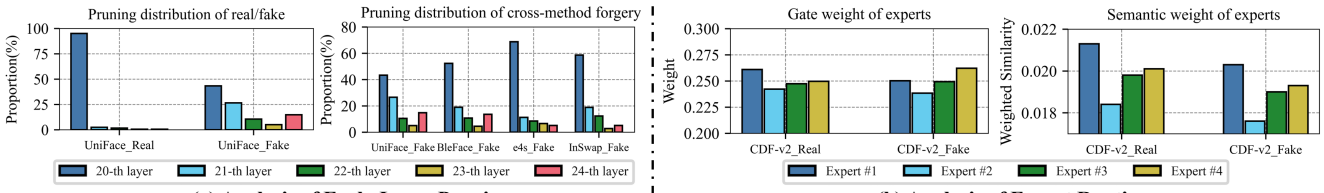
num	CDF.	Fsh.	Avg.
2	94.9	90.5	92.7
4	**96.0**	**91.2**	**93.6**
6	95.3	90.9	93.1
8	95.0	91.0	93.0

(e) **Expert Number**. The larger expert number will not be better.

param	CDF.	Fsh.	Avg.
0.1	**96.0**	**91.2**	**93.6**
0.5	95.4	90.7	93.1
1.0	94.6	90.3	92.5

(f) **Loss balancing**. Rank loss balanced by 0.1 is better.

Table 8. **DFD-HR detailed ablation experiments** with CLIP ViT-L/14 trained on FF++_c23 [63]. We report video-level AUC (%) on CDF-v2 and Faceshifter. If not specified, default settings are marked in blue .



(a) Analysis of Early Layer Pruning **(b) Analysis of Expert Routing**

Figure 5. **Analysis of Hierarchical Routing.** Sub-figure (a) shows that real samples predominantly cut off early while fake ones tend to flow deeper and differ across methods. Sub-figure (b) indicates that Expert #4 and Expert #1 act as the forgery expert and semantic expert, respectively. For all datasets, these two experts dominate the learning between real and fake samples.

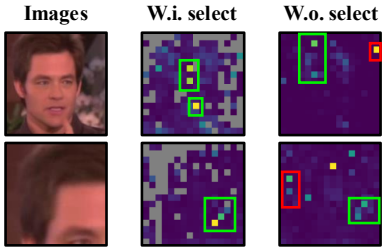


Figure 6. **Visualization of Token Selection of multi-scale images.** Face images shown in figure are from Celeb-DF-v2 [40].

monotonically decreasing shape. In particular, for pruning distribution comparison between real and fake samples, real distribution exhibits a pronounced peak, while the fake one is more uniformly spread. This indicates that real samples predominantly cut off early while fake ones tend to flow deeper and differ across methods, thus providing strong evidence for the necessity of Early Layer Tuning (Fig. 2).

Analysis of Token-level Expert Routing. We further analyze the weight distribution of gate network for all selected tokens (Fig. 5 (b)). Our preliminary observation is that the distributions are largely stable, enabling each expert to specialize without becoming a bottleneck. In addition, an interesting phenomenon is that for all datasets, Expert #1 and Expert #4 dominate the learning between real and fake samples. We further multiple the gate weights of selected tokens with corresponding cosine similarity multiplied by global semantic query h^{cls}_{global}. For all datasets, Expert #1 dominates the learning of semantic information, and we observe no statistically significant difference between the semantic distributions of real and fake samples. Consequently, Expert #4 and Expert #1 function as forgery and semantic expert, respectively, thereby decoupling forgery learning from semantic representation (Fig. 2(a)).

Visualization of Token Selection. To gain more insights into Token Selection, we visualize attention maps of multi-scale images (Fig. 6). We observe that spurious tokens (red region) are filtered by Token Selection with Spearman rank loss for better forgery learning (green region).

5. Conclusion

In this paper, we start from an import question: although only a few parameters are tuned, have existing PEFT-based detectors truly exploited the most informative ones while eliminating redundant parameters for better generalization? Since latent features across layers capture different semantic abstractions and tokens within the same layer convey varied forgery cues, we move beyond standard PEFT by integrating both layer- and token-level routing to maximize representational synergy. Specifically, at the layer level, we introduce "Early Layer Pruning", an adaptive truncation mechanism that enables the model to adaptively learn distinct forward depths for different instances. At the token level, 'Token Selection" is guided by the Spearman rank loss to filter tokens irrelevant to forgery learning. Furthermore, a unified MoE architecture is applied that reduces the potential model's overfitting to specific forgery types. Extensive results demonstrate the effectiveness of our designs and superior performance over existing SoTAs.

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